

Vehicle Parking App - V2 (Project Report)

Student Details:

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Greetings everyone,

I'm Sheen Gauhar from Dhanbad, Jharkhand, currently 19 years old and pursuing BS in Data Science and Applications from IIT Madras, standalone. I'm grateful to be part of this prestigious institute and this project, which has introduced me to valuable new learnings.

Problem Statement:

This project implements a Vehicle Parking Management System (VPMS) that enables smooth interaction between Admins and Users through a unified web interface and dedicated dashboards.

It provides real-time parking availability, spot reservation and cancellation, and automated tracking of active reservations, all while ensuring secure **token-based authentication** and efficient system operations powered by **background jobs**.

Key Features:

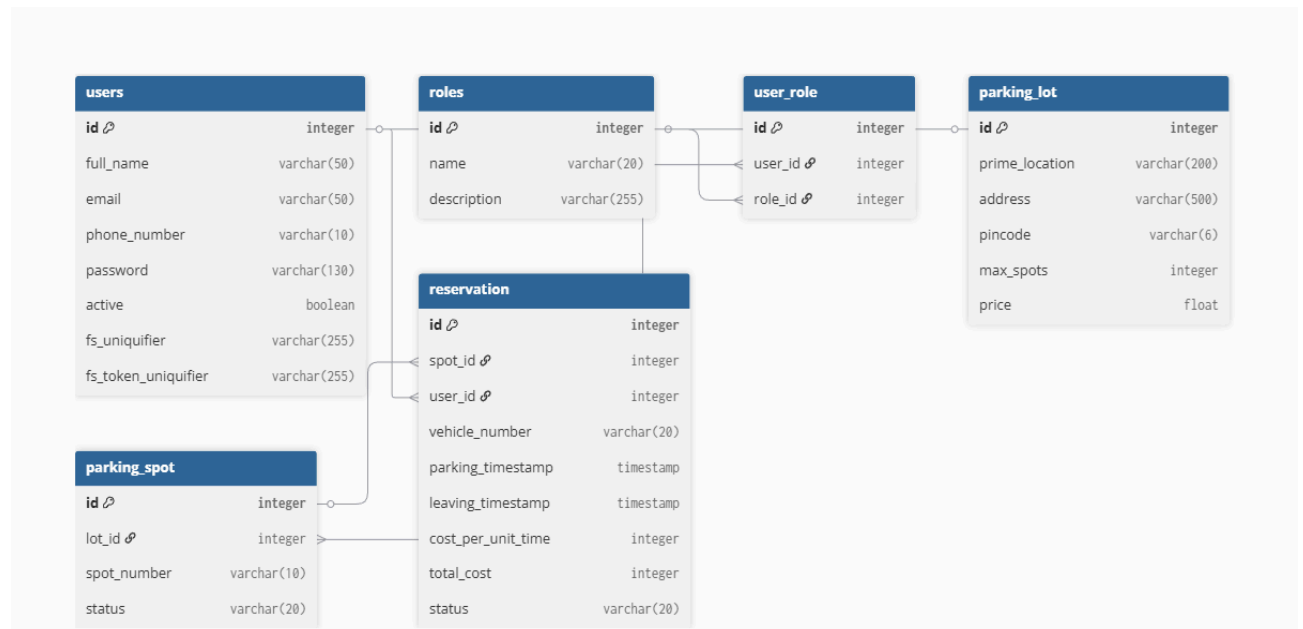
- **Admin Dashboard:** Manage parking lots, auto-generated spots, and monitor real-time user activity with clear visual summaries.
- **User Dashboard:** Book and cancel parking spot reservations, track active bookings, view complete reservation history via summaries, and download CSV for past reservations.
- **Authentication & Role Management:** Secure login with token-based authentication, offering separate interfaces and permissions for Admin and User roles.
- **Summary Charts:** Dynamic data visualization displaying parking usage trends for users and reservation/income analytics for admins.
- **Responsive UI:** Built using Vue 3, HTML, and CSS, ensuring a smooth and consistent experience across desktops, tablets, and mobile devices.

Technologies Used:

Technology	Purpose
Flask (Python)	Backend web framework for building REST APIs and serving the application.
Vue.js	Lightweight frontend for building interactive and responsive UI.
SQLite	Lightweight, file-based database for persistent storage.

SQLAlchemy	ORM used for database table mapping and operations.
Redis	Used as an in-memory store for caching and as a Celery message broker.
celery	Handles asynchronous and scheduled background tasks(daily reminders, monthly reports, CSV exports).
Flask Security	Provides secure token-based authentication and role-based access control
Flask-CORS	Enables frontend-backend communication across origins.

ER Diagram:



API Design:

Base URL: /api/

Endpoint	Method	Role	Functionality
/login	POST	All	Authenticate user and return auth token
/logout	POST	Any authenticated	Logs out the current user

/register	POST	All	Creates a new user account
/check-email	POST	All	Checks if an email is already registered
/parking_lots	POST	Admin	Create a new parking lot + auto-create spots
/parking_lots/<lot_id>	PUT	Admin	Update parking lot details & spot structure
/parking_lots/<lot_id>	Delete	Admin	Deletes a parking lot (only if no occupied spots)
/admin/users	GET	Admin	View all registered users
/spot/<spot_id>/user_info	GET	Admin	View which user occupies a parking spot
/admin/summary/revenue	GET	Admin	Get total revenue summary (cached)
/admin/summary/spot_status	GET	Admin	Get occupied/available spot count per lot
/profile	GET	User/Admin	Get current user's profile
/profile/edit	PUT	User/Admin	Edit profile (name, phone)
/user/reservations	GET	User	Get all reservations of logged-in user
/parking_lots	GET	User	List lots with available spots only
/parking_lots/<lot_id>/next_spot	GET	User	Get nearest available spot in a lot
/reservations	POST	User	Create a new reservation (booking a spot)
/reservations/<reservation_id>/release	POST	User	Release spot + calculate cost
/user/clear_history	DELETE	User	Delete past (left) reservations

/user/summary/usage_per_lot	GET	User	Get usage statistics for each spot (user-specific)
/export_csv	POST	User	Export user's reservation history via Celery Email

AI/LLM Used percentage and extra details: Overall - 34%; Some routes logic; styling & Layout; Logic for summary of available vs occupied spot, revenue & for spots used by user, debugging backend jobs error.

For Project Video, [click here](#)